

Trap Energy Distribution in Polymeric Insulating Materials through Surface Potential Decay Method

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ABSTRACT

A model based on surface potential decay (SPD) measurement was developed to study the trap energy distribution of hole-type and electron-type traps in materials by considering two distinct energy levels (i.e., shallow and deep traps) and the detrapping process. An approximated Dirac function $\delta((E-E_m)/kT)$ was also introduced to obtain a consistent unit of trap density. Besides, the space charge distributions (such as uniform, linear and exponent distribution) in positively and negatively charged materials were also taken into account. In order to verify our proposed model, the different SPD curves of low density polyethylene (LDPE), LDPE/zinc oxide (ZnO) microcomposite and nanocomposite were analyzed. The results show that the proposed model is effective to obtain the trap energy distribution in materials. The SPD curves in positively and negatively charged samples show a fast decay followed by a slow decay, which is agreed with our assumption of shallow and deep traps. Two peaks are clearly observed from the trap energy distribution. The trap energy distributions of hole-type and electron-type traps in these materials are distinct from each other. This implies some essential nature of hole-type and electron-type traps (i.e., intra-chain character for hole-type traps and inter-chain character for electron-type traps). In addition, compared to LDPE and LDPE/ZnO microcomposite, the decreasing energy level and the increasing trap density of shallow traps as well as the decreasing trap density of deep traps in LDPE/ZnO nanocomposite may be attributed to the effect of the interaction zone surrounding nanoparticles.

Index Terms — LDPE, nanocomposite, trap energy distribution, surface potential decay, morphology.

1 INTRODUCTION

POLYMERIC insulating materials have been widely used in the electrical and electronic applications, but their electrical performances especially in high voltage dc electric field are greatly affected by the space charge accumulation. A considerable space charge accumulation may cause a severe distortion in the local electric field, resulting in the local material being overstressed even the degradation or aging as well as premature failure. Thus, the space charge accumulation in polymeric insulating materials has drawn a great attention in recent decades [1, 2]. It has been proved that the space charge formation is closely related to the charge trapping and detrapping characteristics [3], and the trap distribution is then usually used to investigate the space charge property of polymeric insulating materials.

The trap distribution or trap parameters can be measured by various measurement methods, such as thermally stimulated current (TSC) [4], isothermal relaxation current (IRC) [5], pulsed electroacoustic (PEA) [3, 6] and surface potential decay (SPD) [7,8]. Both simulation and experiment have shown that the trapping and detrapping characteristics of hole and electron carriers are normally different from each other [2, 9]. Generally, both TSC and IRC methods measure the relaxation current, whereas the PEA method measures the net charges. The relaxation current or net charges consist of hole and electron charge carriers. Consequently, the above methods cannot separate the electron-type traps from the hole-type ones. In comparison, the SPD method provides an effective way to identify these two types of traps by using different charging polarity.

Although the SPD phenomenon can be attributed to different mechanisms, e.g. surface conduction [10], gas neutralization [11] and polarization processes [12], the detrapping kinetics of charge carriers play an important role in

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contributing to the SPD phenomenon if the charge carriers' transit time is shorter than their mean detrapping time particularly under high electric field [7,8,13,14]. The potential decay rate can then be explained from the aspect of energy. The detrapping kinetics suggests that the charge carriers in shallower traps release following by those in deep traps. The trap distribution was obtained by Watson according to the IRC theory and SPD experiment without any pre-knowledge on the distribution of charge trap states [7], but the trap density was just described by relative values. An improved model was proposed to solve the relative values of trap density, assuming that the current induced by the detrapping carriers is equal to that obtained from the SPD curve [8]. Considering the relationship between surface potential and space charge profile, the trap distribution was redefined, not only directly reflecting the release of charge carriers in trap centers, but also revealing the distinct nature of electron and hole traps. However, in this method, the approximated special function $G_n(E, t) = (kT/t)\delta(E-E_m)$ may cause an inconsistent unit comparing with special function $G_n(E, t) = e_n \exp(-e_n t)$. In addition, only single trap was taken into account. By using a double exponential fitting, the shallow and deep traps were discussed based on SPD results [13], but the origin of these traps was not clearly illustrated. Moreover, the influence of spatial distribution of charge carriers in the sample on the trap distribution was analyzed, and the application of a special function $G_n(E, t) = (1/t)\delta(E_T, t)$ has avoided the existence of inconsistent unit used in the trap distribution, but the independent variable of Dirac function $\delta(E_T, t)$ was not given out clearly [14].

In this paper, a quantitative analysis on the trap distribution of polymeric insulating materials has been carried out by SPD measurement combined with the assumed space charge distributions, two trap levels and IRC theory. The influences of charge uniform, linear and exponent distribution in the corona charged samples are discussed in details. A SPD experimental setup is used to measure the SPD characteristics of low density polyethylene (LDPE), LDPE/zinc oxide (ZnO) microcomposite and nanocomposite. Based on our proposed model, the trap distributions of hole and electron traps of these materials are then obtained. The results show that the shallow/deep trap distributions of hole-type traps are distinct from those of electron-type ones.

2 THEORETICAL MODEL OF TRAP ENERGY DISTRIBUTION

2.1 MODEL OF SURFACE POTENTIAL DECAY

For SPD measurement, the polymeric insulating material is placed on a metal platform. The bottom surface of the insulating material is grounded. The top surface is charged by a corona discharge system, and a 1~2 μm thin layer of homocharges are then formed. At the same time, heterocharge at the bottom surface will be induced by Schottky or Fowler-Nordheim effect. In other words, the double injection phenomenon takes places in the corona charged material. This has been verified by the space charge measurements on the corona charged material [15,16]. To simplify the study on the

relationship between the surface potential and the charges trapped in the corona charged sample, the same injection depth of homocharges and heterocharges is usually used [8].

Considering the space charge distribution in one dimension and the double injection phenomenon, the space charge distributed in a negatively corona charged sample can be divided into three layers (namely, the bottom surface (0~ l), the bulk (l ~ $L-l$) and the top surface ($L-l$ ~ L)) as shown in Figures 1a, where l is the maximum injection depth and L is the thickness of sample.

Assume that the density of space charge distributed in each layer is expressed as ρ_b , ρ , ρ_t , respectively, and the surface potential (U_s) is then calculated as:

$$U_s = \frac{1}{\varepsilon_0 \varepsilon_r} \int_0^L x \rho_x dx$$

$$= \frac{1}{\varepsilon_0 \varepsilon_r} \left(\int_0^l x \rho_b dx + \int_l^{L-l} x \rho dx + \int_{L-l}^L x \rho_t dx \right) \quad (1)$$

where ε_0 is the vacuum permittivity, ε_r is the relative permittivity and ρ_x is the charge density at position x .

Accordingly, the surface potential is closely related to the space charge distribution. After the sample is corona charged in

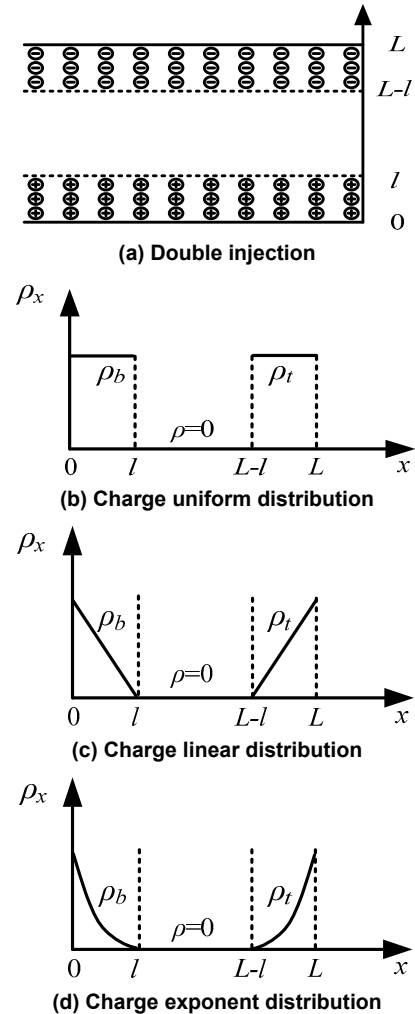


Figure 1. Schematic diagram of space charge distribution in negatively corona charged polymeric insulating material.

a short time, most of the injected charges are accumulated on the sample's surface whereas almost no space charges exist in the sample's bulk, namely $\rho=0$. So the space charge distributions near the top and bottom surface play an important role in the establishment of surface potential. In this paper, three different kinds of space charge distribution (i.e., uniform, linear and exponent distribution) are taken into account, as shown in Figure 1. The relationships between the surface potential and the charge density under the assumed space charge distributions are discussed as follows.

The charge uniform distribution is usually used to obtain the trap energy distribution in the corona charged material [8, 13]. The charge density in the top and bottom layers is independent of injection depth as shown in Figure 1b. In other words, the charge density in any position (x) of the top and bottom layers is equal in amount to the surface charge density (σ) on the top surface ($x=L$), i.e:

$$\rho_b = \rho_t = \sigma \quad (2)$$

Substituting Equation (2) into Equation (1) and considering the condition of $l \ll L$, the relationship between the surface potential and the surface charge density on the top surface can be simplified as follows:

$$U_s = \frac{Ll\sigma}{\epsilon_0\epsilon_r} \quad (3)$$

For charge linear distribution, the relationship between the charge density in the top (or bottom) layer and the position x is assumed to be linear. As shown in Figure 1c, the charge density in the top layer is increased linearly with the increasing x , whereas that in bottom layer is decreased linearly with the increasing x . Accordingly, the charge density in the top and bottom layers can be denoted as:

$$\rho_b = \frac{\sigma_b}{l}(l-x) \quad (4a)$$

$$\rho_t = \frac{\sigma}{l}(x-L+l) \quad (4b)$$

where σ_b is the surface charge density on the bottom surface ($x=0$). Considering the characteristics of charge uniform distribution, it is assumed that, for charge linear distribution, the top and bottom surfaces have the same surface charge density, i.e., $\sigma_b = \sigma$. Consequently, according to Equation (1) and Equation (4) as well as $l \ll L$, the surface potential can be rewritten as:

$$U_s = \frac{1}{2} \times \frac{Ll\sigma}{\epsilon_0\epsilon_r} \quad (5)$$

For charge exponent distribution as shown in Figure 1d, the charge density in the top and bottom layers changes exponentially with the increasing x . The specific relationships between the charge density and the position x in the top and bottom layers are defined as follows, respectively,

$$\rho_b = \frac{\sigma}{1-e} \left(1 - \exp\left(\frac{l-x}{l}\right) \right) \quad (6a)$$

$$\rho_t = \frac{\sigma}{1-e} \left(1 - \exp\left(\frac{x-L+l}{l}\right) \right) \quad (6b)$$

Combining Equations (1) and (6) as well as the condition of $l \ll L$, the surface potential now can be described by,

$$U_s = \frac{e-2}{e-1} \times \frac{Ll\sigma}{\epsilon_0\epsilon_r} \quad (7)$$

From the above, we can learn that the surface potential is mainly determined by the homocharges accumulated on the top surface. The above analysis is also applicable to the positively corona charged sample. The major difference from the negatively corona charged sample is that the positive charges are accumulated on the top surface whereas the negative charges induced by Schottky or Fowler-Nordheim effect are formed on the bottom surface.

2.2 MODEL OF SHALLOW AND DEEP TRAPS

The charges accumulated on the top surface of the corona charged sample will experience trapping and detrapping process to arrive the ground electrode when the transit time is shorter than their mean detrapping time. The charge trapping and detrapping characteristics are greatly affected by the trap energy distribution in the sample. A semicrystalline polymeric material may have some discrete energy levels or even continuous trap energy distribution [3, 6]. In this model, we assume that the traps only have two energy levels, which are distinguished as shallow traps and deep traps [3, 6, 13, 14]. In addition, the traps are assumed to be uniformly distributed in the sample. The trapping and detrapping processes of charge carriers between shallow traps and deep traps are not considered, and the charge carriers detrapped from trap centers are assumed to be not captured by traps again [8, 13].

After being negatively corona charged, the sample begins to experience detrapping process. The detrapping process may be attributed mainly to the thermal detrapping [13, 14] when the applied electric field is not high enough to activate the impact ionization and tunneling [17]. The changing rate of trapped charge density caused by the thermal detrapping can be defined by:

$$\frac{dn_{e1}}{dt} = -r_{de1}n_{e1} \quad (8a)$$

$$\frac{dn_{e2}}{dt} = -r_{de2}n_{e2} \quad (8b)$$

where n_{e1} and n_{e2} are the density of trapped charges in shallow and deep traps, respectively. r_{de1} and r_{de2} are the detrapping probability of trapped charges in shallow and deep traps, respectively. If the trap energy level levels of insulating materials are distributed throughout the band gap, the detrapping probabilities (r_{de1} and r_{de2}) are closely related to the energy level and they can be uniformly described as [3, 5-8, 3, 14]:

$$r_{dei}(E, T) = v \exp\left(\frac{E - E_c}{kT}\right) \quad (9)$$

where ν is the attempt to escape frequency, E is the energy level, E_c is the energy of conduction band edge, k is the Boltzmann's constant and T is the Kelvin temperature. $i=1,2$.

By integrating Equations 8a and 8b, respectively, the density of trapped charges in shallow and deep traps can be obtained, as follows:

$$n_{e1}(t) = n_{e10} \exp(-r_{de1}t) \quad (10a)$$

$$n_{e2}(t) = n_{e20} \exp(-r_{de2}t) \quad (10b)$$

where n_{e10} and n_{e20} are the initial density of trapped charges in shallow and deep traps, respectively. Consequently, the density of total trapped charges (n_e) in both shallow and deep traps can be represented by $n_e = n_{e1} + n_{e2}$. Moreover, it is also directly concerned with the surface charge density (σ) on the top surface, described by:

$$n_e = \sigma/q \quad (11)$$

2.3 THEORY OF ISOTHERMAL RELAXATION CURRENT

During the charging process, the IRC theory shows that part of the traps above the equilibrium Fermi level E_{f0} (electron traps) or parts of those below E_{f0} (hole traps) will be occupied by injected electrons or holes. After removal of the applied electric field, the electrons or holes captured by the electron traps or hole traps will be thermally emitted into the conduction band or valence band according to the above assumption. Taking negatively corona charging for example, the rate of all the electrons emitted from traps can be described as [5]:

$$\frac{dn_e}{dt} = \int_{E_i}^{E_c} f_0(E) N(E) r_{de} \exp(-r_{de}t) dE \quad (12)$$

where $f_0(E)$ is the initial occupancy of the traps by electrons. $N(E)$ is trap density at the energy level of E . E_i is the maximum energy of electron traps.

A special function $G_e(E, t)$ is introduced to described the weight contribution of any energy level of electron traps to the relaxation current and is given by:

$$G_e(E, t) = r_{de}(E, T) \exp(-r_{de}(E, T)t) \quad (13)$$

For a fixed t , the plot of $G_e(E, t)$ presents an asymmetrical bell-shape with a maximum value at $E=E_m$ and a half width of $3kT$. According to Equation (13) and the plot characteristic of $G_e(E, t)$ as described in [5], the energy level of electron traps (E_e) can be obtained as:

$$E_e = E_c - E_m = kT \ln(\nu t) \quad (14)$$

Due to the similar plot, the special function $G_e(E, t)$ may be approximated by the Dirac function, which is defined by:

$$\begin{aligned} G_e(E, t) &= r_{de}(E, T) \exp(-r_{de}(E, T)t) \\ &= B \delta\left(\frac{E - E_m}{kT}\right) \end{aligned} \quad (15)$$

where B is a normalization constant.

By integrating both sides of Equation (15) with respect to energy level from E_i to E_c , the following relationship can then be obtained from:

$$\begin{aligned} B \cdot kT &= \int_{E_i}^{E_c} r_{de}(E, T) \exp(-r_{de}(E, T)t) dE \\ &= (kT/t)(1 - \exp(-\nu t)) \end{aligned} \quad (16)$$

The attempt to escape frequency ν of trapped charges is usually of the order of 10^{11} to 10^{13} s^{-1} , and then the $\exp(-\nu t)$ is much smaller than 1 for realistic values of t . As a result, the normalization constant B can be expressed by:

$$B \approx 1/t \quad (17)$$

The special function $G_e(E, t)$ is then rewritten by:

$$G_e(E, t) = \delta\left(\frac{E - E_m}{kT}\right) / t \quad (18)$$

By substituting Equation (18) into Equation (12), the changing rate of trapped charge density is then rewritten as:

$$\frac{dn_e}{dt} = \frac{f_0(E) N_e(E)}{t} \quad (19)$$

According to Equation (10) and Equation (19), the trap density of shallow and deep traps is obtained, respectively, i.e.:

$$N_{e1}(E_{i1}) = \frac{n_{e10} r_{de1} t \exp(-r_{de1}t)}{f_0(E_{i1})} \quad (20a)$$

$$N_{e2}(E_{i2}) = \frac{n_{e20} r_{de2} t \exp(-r_{de2}t)}{f_0(E_{i2})} \quad (20b)$$

where $N_{e1}(E_{i1})$ and $N_{e2}(E_{i2})$ are trap density at the energy level of E_{i1} and E_{i2} for shallow traps and deep traps, respectively.

The trap density of electron-type shallow and deep trap traps is closely related to the initial density and the detrapping probability of trapped charges as well as the initial occupancy of the electron traps. The former two can be obtained from the SPD experimental data according to the proposed model, whereas the third one (i.e., the initial occupancy of the electron traps) is usually assumed to be 1/2. All the above analysis can also be used to hole traps, but the only difference is that the hole traps are below the Fermi level. Compared with electron traps, the hole traps is referred to the energy of valence band edge (E_v).

3 SURFACE POTENTIAL DECAY MEASUREMENT AND CALCULATION OF TRAP ENERGY DISTRIBUTION

3.1 EXPERIMENTAL DETAILS

A specific experimental setup located in a climatic chamber was established to measure the surface potential and was composed of a corona charge system and a measurement system, as shown in Figure 2. A sample was first charged for 3 min by the corona charge system with ions of either sign.

The corona charge system consists of a high-voltage needle electrode, a mesh grid and a grounded electrode. The needle electrode is used to generate a corona discharge, and the mesh grid is applied to produce reasonably uniform field in the gap between the mesh grid and the grounded electrode. The needle electrode and mesh grid are applied with ± 11 kV and ± 3 kV dc high voltage, respectively. After being corona charged, the sample was quickly moved to the measurement system via a rotatable platform for measuring the surface potential characteristics. The measurement system consists of a Kelvin type probe, a noncontact electrostatic voltmeter and a computer. The model of the electrostatic voltmeter is Trek P0865 with a resolution of 1V. The SPD measurement for each specimen was performed for three times and the SPD characteristics show good repeatability. All experiments were carried out at 20 °C with relative humidity of about 40%.

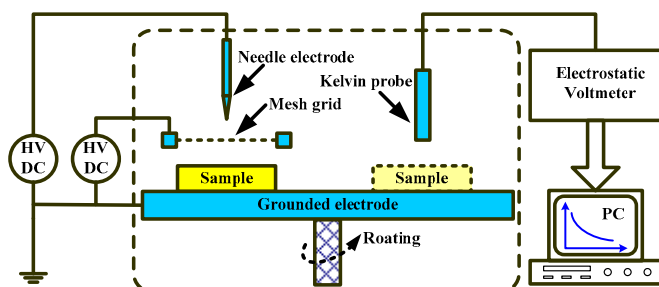


Figure 2. Surface potential measurement system.

LDPE, LDPE/ZnO nanocomposite (LZN) and LDPE/ZnO micro-composite (LZM) were prepared through a hot-pressing process. The average particle sizes of nano- and micro- ZnO particles are 40 nm and 3 μm , respectively. The ZnO particles are surface treated using silane coupling agent (KH550). The nano- and micro- filler loading were 20 wt%, respectively. The selected powder was blended with LDPE by an open mill at 110 °C for 40 min. After blending process, the hot-pressing process was carried out to obtain the specimen with a thickness of 200 μm . Before the SPD measurement, all the specimens were cleaned with anhydrous alcohol and then dried at 60 °C for 3 h to eliminate the humidity and the residual charges.

Figure 3 shows a scanning electron microscope (SEM) image of the 20 wt% LDPE/ZnO nanocomposite. The ZnO nanoparticles are shown as white spheres. Almost no agglomeration of ZnO nanoparticles can be observed. This means that the ZnO nanoparticles are uniformly distributed in LDPE.

3.2 EXPERIMENTAL RESULTS OF SURFACE POTENTIAL DECAY

Typical normalized SPD characteristics of positively and negatively corona charged LDPE, LZM and LZN are shown in Figure 4. The normalized surface potential (U_{st}^*) is obtained according to the initial surface potential (U_{s0}) and the surface potential (U_{st}) at any time (t) [8]. It can be presented by $U_{st}^* = U_{st}/U_{s0}$. The initial surface potential is +2137/-2128 V for LDPE, +1984/-1923 V for LZM and +2019/-1856 V for LZN.

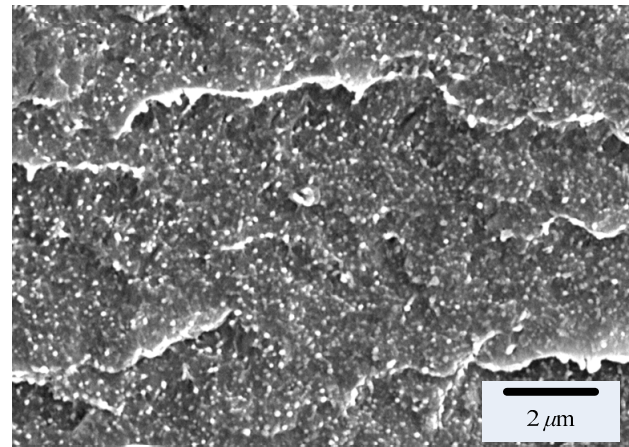


Figure 3. SEM image of the 20 wt% LDPE/ ZnO nanocomposite.

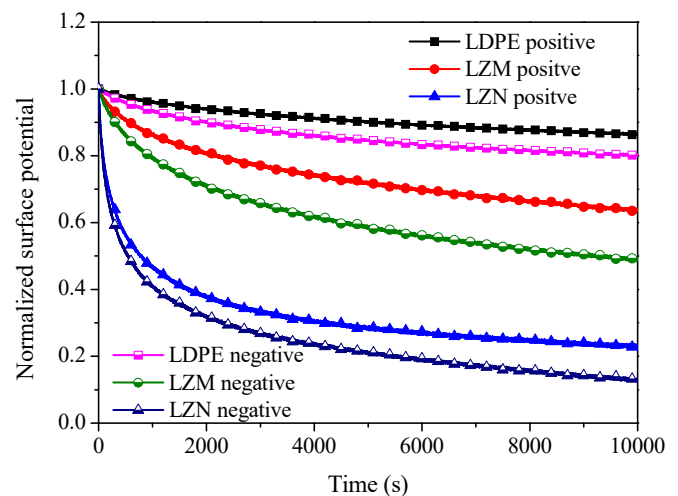


Figure 4. Surface potential decay characteristics of LDPE, LZM and LZN.

It can be seen from Figure 4 that the positive and negative SPD curves of all specimen behave a similar tendency. Namely, the surface potential decays faster at the initial time but gradually slows down with the passage of time whether LDPE is charged positively or negatively. Similar charge decay phenomena were observed from PEA and SPD measurement [3, 6, 13, 14], and are attributed to the two energy levels (i.e., shallow and deep traps) existed in materials. This fact is in consistent with the assumption described in Section 2.

Figure 4 also show that the positive and negative SPD curves of each specimen are distinct from each other. The SPD curve of positively charged specimen decays slower than that of negatively charged one. Although ions are the charge carriers that impinge onto the sample, studies aimed at explaining the surface potential build up of additive-free polymers or nanocomposite have shown that, charge transport is dominated by holes and electrons for the positively and negatively charged specimens, respectively [8,13,14,18,19]. So the distinct positive and negative SPD curves may imply the different energy trap distribution of hole-type and electron-type traps in the materials. Experimental results from SPD measurement [8,13, 14] have shown that traps in LDPE and its composites have continuous energy distribution. Thus, in

this paper, traps are considered to have continuous energy distribution. From Figure 4, we can infer that the hole-type traps may be deeper than that of electron-type ones for each sample. Compared with LDPE, the SPD curves in LZM and LZN decay much faster after either positively or negatively corona charging. It can be deduced that the hole-type or electron-type traps in LZM and LZN are much shallower than that in LDPE, and less energy is needed for the trapped charges in the first two materials to be motivated.

3.3 CALCULATION OF PARAMETERS USED IN THE PROPOSED MODEL

The trap energy distribution of the studied materials can be obtained according to the calculation flowchart as shown in Figure 5. For each assumption of space charge distribution (namely, uniform, linear and exponent distribution), the surface charge density (σ) on the top surface can be obtained from the SPD curves by Equation (3), (5) or (7), respectively. Injection depth l is set $2\ \mu\text{m}$ as presented in [8]. Substituting the values of σ into Equation (11), the total charge density (n_e) in shallow and deep traps is then calculated. Thereafter, the data of n_e is fitted by Equation (10) and the fitting results are used to compute the trap density of shallow and deep traps. Whereas the energy level of shallow and deep traps can be obtained directly from Equation (14).

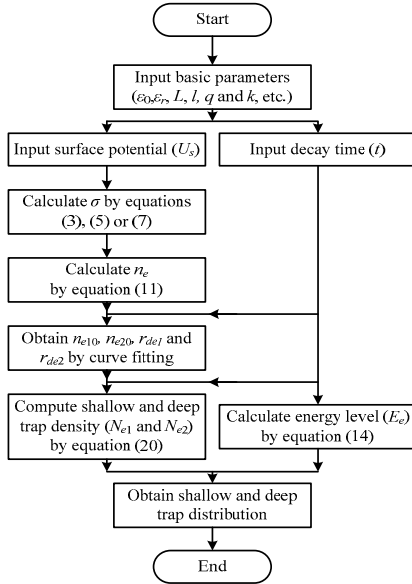


Figure 5. Flowchart for calculating the shallow and deep trap distribution.

To investigate the influence of the space charge distribution in the corona charged materials on the trap energy distribution, LDPE charged by positive and negative polarity is taken for example. By following the flowchart in Figure 5, the parameters (n_{e1} , n_{e2} , r_{de1} and r_{de2}) which play an important role in determining the trap energy distribution are obtained, as shown in Table 1.

It can be found from Table 1 that, for positively corona charged LDPE, the initial density of trapped charges in shallow traps (n_{e10}) is much lower than that in deep traps (n_{e20}) for each kind of the assumed charge distributions. This means

that more charges are trapped in deep traps than shallow traps. Besides, the values of n_{e10} and n_{e20} are closely related to the space charge distribution type in LDPE and are nearly increased when the space charge distribution is changed from the uniform distribution to the exponent distribution. This indicates the space charge distribution can affect the trapped charge density to some extent. In contrast, the thermal detrapping probabilities (r_{de1} and r_{de2}) are constant regardless of the space charge distribution types. The value of r_{de1} is two orders in magnitude greater than that of r_{de2} , which suggests that charges are easier to move out from shallow traps than deep traps. A similar variation tendency is also found for negatively corona charged LDPE.

Table 1. Parameters derived from fitting data for positively (P) and negatively (N) corona charged LDPE

Distribution type	$n_{e10}(\text{m}^{-3})$	$n_{e20}(\text{m}^{-3})$	$r_{de1}(\text{s}^{-1})$	$r_{de2}(\text{s}^{-1})$
Uniform (P)	9.12×10^{19}	1.01×10^{21}	3.4×10^{-4}	5.3×10^{-6}
Linear (P)	1.82×10^{20}	2.03×10^{21}	3.4×10^{-4}	5.3×10^{-6}
Exponent (P)	2.18×10^{20}	2.42×10^{21}	3.4×10^{-4}	5.3×10^{-6}
Uniform (N)	1.52×10^{20}	1.01×10^{21}	4.2×10^{-4}	6.8×10^{-6}
Linear (N)	3.04×10^{20}	2.02×10^{21}	4.2×10^{-4}	6.8×10^{-6}
Exponent (N)	3.64×10^{20}	2.39×10^{21}	4.2×10^{-4}	6.8×10^{-6}

According to the fitting parameters as shown in Table 1 and our proposed model, the trap energy distributions in positively and negatively corona charged LDPE are obtained. As the ones under all charge distributed assumptions behave similar to each other, only the trap energy distributions based on charge uniform distribution are presented here as shown in Figure 6. It can be seen that the unit of trap density obtained by the proposed model is m^{-3} instead of $\text{eV}^{-1}\text{m}^{-3}$ as described in [8]. This indicates that the consistent unit of trap density with that presented in [5,6] is obtained by using the proposed model. Furthermore, two peaks are clearly observed from Figure 6 regardless of the positively or negatively charged LDPE. In each curve, the left peak with a lower energy level represents the shallow traps whereas the right one with a higher energy level represents the deep traps. This is in accordance with the assumption of shallow and deep traps

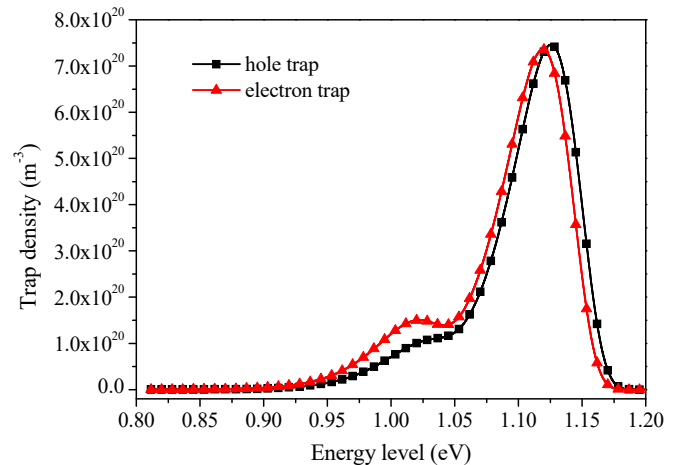


Figure 6. Trap energy trap distribution of positively and negatively charged LDPE based on charge uniform distribution.

proposed in the paper. In order to further study the trap energy distribution of shallow and deep traps, the total trap energy distributions as shown in Figure 6 are divided into two single peaks. Figure 7 shows the trap energy distribution of shallow and deep traps of positively and negatively corona charged LDPE based on charge uniform distribution. In addition, the cases of charge linear and exponent distribution are also illustrated in Figure 7.

It can be seen that the trap energy distributions of hole-type or electron-type shallow traps in Figure 7a are similar to each other for different space charge distributions, and the trap centers are not varied with the space charge distribution types. The difference between each other lies in the magnitude of trap density. Similarly, the deep trap distributions as shown in Figure 7b can also be analyzed in this way. Consequently, all of the assumed space charge distribution (namely, uniform, linear and exponent distribution) can be used to analyze the trap distribution of the insulating material.

From Figure 7a and 7b, it can be observed that the trap energy distributions of hole-type shallow and deep traps are quite different from those of electron-type ones based on the same charge distribution. For the shallow trap distributions as shown in Figure 7a, the peak density of hole-type shallow

traps is located at 1.02 eV whereas it is located at 1.01 eV for electron-type ones. For the deep trap distributions as shown in Figure 7b, the corresponding peak density is located at 1.13 eV for hole-type deep traps and it is located 1.12 eV for electron-type ones. These facts demonstrate that the mean energy level of hole-type traps is much higher than that of electron-type traps. In addition, for the hole-type traps, the shallow trap density is smaller than the deep trap density, so as to the trap density of electron-type traps. The results indicate that deep traps account for a large proportion of LDPE. Furthermore, the trap density of hole-type shallow traps is smaller than that of electron-type ones. Comparing with electron-type traps, the higher shallow and deep energy levels as well as the smaller shallow trap density for hole-type traps may contribute to the fact that the positive SPD curves decays slower than the negative ones.

3.4 HOLE-TYPE AND ELECTRON-TYPE TRAP ENERGY DISTRIBUTIONS IN NANOCOMPOSITE

Two peaks also can be observed from the trap energy distribution in positively or negatively charged LZM and LZN, compared to that in LDPE as shown in Figure 6. In order to avoid the content redundancy, only the trap energy distributions of hole-type (or electron-type) shallow and deep traps in LZM and LZN are illustrated here as shown in Figure 8 based on the uniform charge distribution. For LZN, the peak density of hole-type shallow and deep trap distribution are located at 0.98 and 1.06 eV, respectively. In contrast, they are located at 0.97 and 1.05 eV for electron-type shallow and deep traps, respectively. The results indicate that the hole-type and electron-type traps in LZN are similar to those in LDPE as mentioned above, so are the ones in LZM. In other words, the hole-type traps are much deeper than the electron-type ones.

It can be seen from the hole-type trap energy distributions as shown in Figure 8a that the introduction of ZnO micro- or nano- particles results in the decrease in the energy level compared to LDPE. Moreover, the energy level of hole-type traps in LZN is much lower than that of LZM. From Figure 8a, it can also be found that the trap density of shallow traps is increased whereas that of deep traps is decreased, when the ZnO nano-particles is filled into LDPE in contrast with composites made with ZnO micro-particles. The electron-type trap energy distributions in Figure 8b behave similar phenomena. It has been proved that the shallower energy level and the greater trap density can lead to the enhancement in charge trapping when LDPE is treated by long-term aging process [20]. In this work, it seems that the special trap energy distribution (namely, the decreasing energy level in shallow and deep traps, the increasing trap density of shallow traps and the decreasing trap density of deep traps) in LZN may contribute to the faster decay rate of the surface potential of LZN than that of LDPE and LZM, as shown in Figure 4.

5 DISCUSSION

The morphology of these three samples may account for the diversity in the trap energy distribution of hole-type and electron-type traps as shown in Figure 8. LDPE is a typical semi-crystalline polymeric material, encompassing both

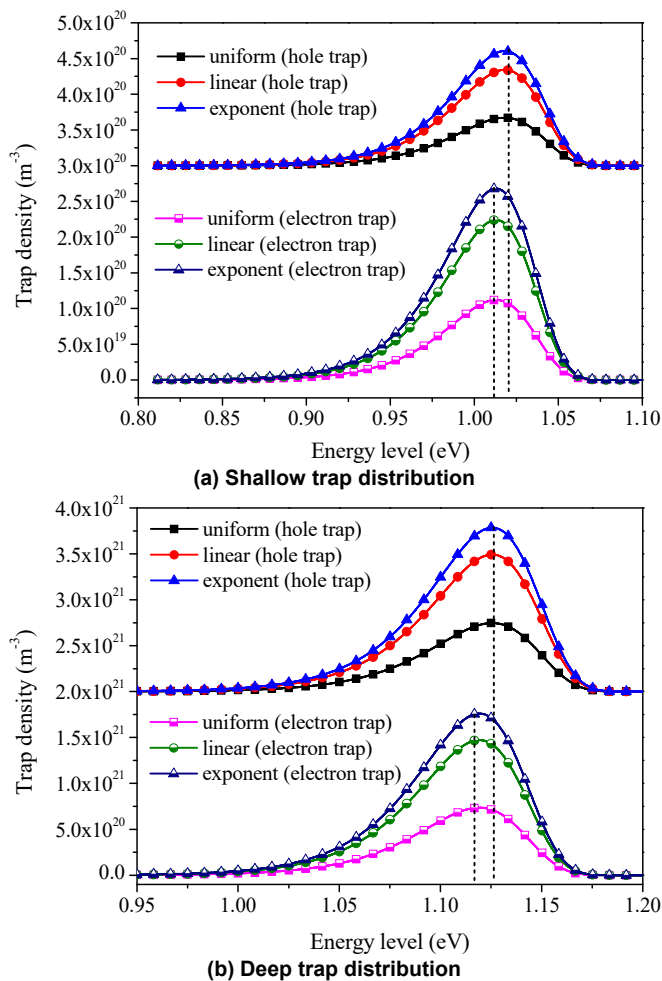


Figure 7. Trap energy distributions of shallow and deep traps in positively and negatively charged LDPE based on charge uniform, linear and exponent distribution.

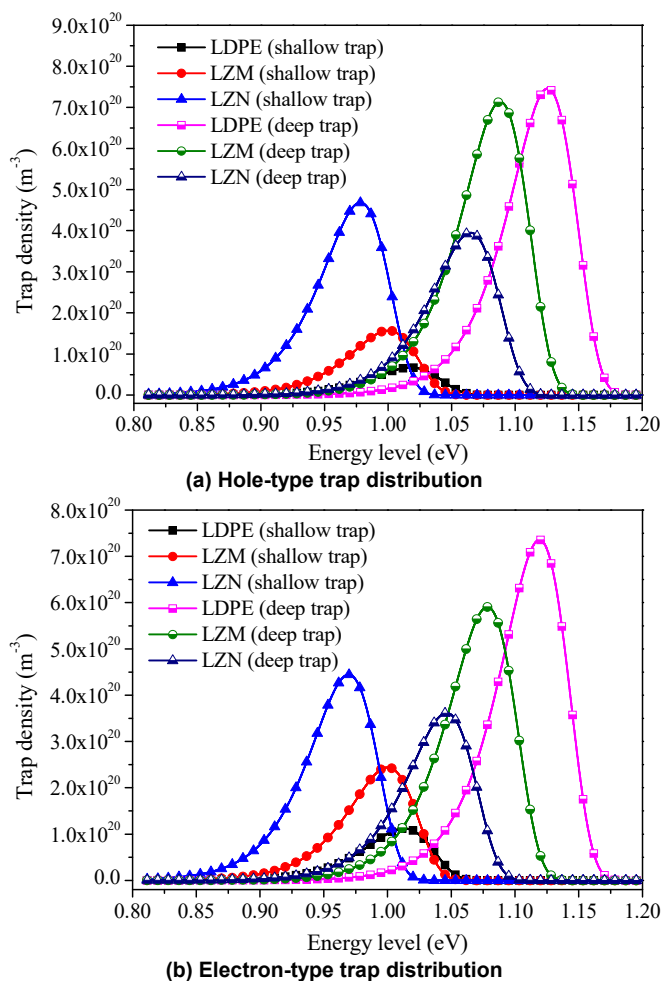


Figure 8. Trap energy distributions of LDPE, LZM and LZN based on charge uniform distribution.

crystalline lamellas and amorphous regions. In the crystalline regions, some chemical defects like double carbon-carbon bonds may exist along the macromolecular chains. In comparison, there exist many shorter linear chains, branched or cross-linked groups with smaller molecules such as byproducts, assistant agent in the amorphous regions. It has been revealed that a number of traps exist in the crystalline regions, the amorphous regions and/or the interfaces between the crystalline and amorphous regions, and the energy level of these traps is about 1 eV [21,22], which is in agreement with the measured mean energy level of 1 ± 0.1 eV as exhibited in Figure 8. These traps, however, may contribute to the trapping or detrapping process of holes and electrons in different ways, due to the distinct mechanisms of hole and electron conduction. Molecular simulation results show that the valence-band states have an intra-chain character whereas the conduction-band edge states have an inter-chain character [9,23]. The intra-chain character of the valence-band states implies a strong bonding force, thus an added hole will be mainly captured and deeply trapped by the chemical defects along the macromolecular chains in the crystalline regions. After removing the applied voltage, the holes trapped require much higher energy to be thermally activated. For the conduction-band edge states, the inter-chain character means

that the binding of an added electron to a single polymer chain is weak and physical. In other words, an added electron will be moved out from the polymer chains to move into a low-density region like the outer wall of a cavity or the interface between the crystalline regions and the amorphous regions. In this low-density region, shallow traps may be formed due to the expanded inter-chain distance and less condensed molecular conformation. The electrons trapped are then easier to be thermally stimulated compared to holes trapped. Consequently, the electron-type traps are much shallower than the hole-type ones in LDPE. This is in accordance with trap energy distribution presented in [8, 13].

As the denser packing of $-\text{CH}_2-$ groups exist in the crystalline regions of LDPE, the micro- and nano- particles will only penetrate the amorphous regions and take effect in these regions [24, 25]. Moreover, ZnO is a typical n-type ohmic semiconductor. Based on the above analysis on trap energy distribution of LDPE, it can be deduced that the electron-type traps are much shallower than the hole-type ones in both LZM and LZN.

The nano-particles filled in a base polymer can alter the polymer morphology structure, such as changes in the crystallization and increases in the interface areas surrounding the nano-particles, as compared to micro-particles [26]. The local physical and chemical properties of the interface areas are different from those of the polymer matrix, and they may change the polymer chain mobility, chain arrangement, crosslink density, free volume, local conductivity and so on [27]. It has been proved by a great number of experimental results [28-30] that the interface areas or interaction zone plays an important role in nanocomposites properties, e.g., deep and shallow traps. The introduction of deep or shallow traps depends on the nanofiller loading. At lower filler loading, the density and/or energy level of the charge carrier trapping states may be increased due to the presence of the interaction zones around nano-particles. TSC results show that the 3 wt% LDPE/zeolite nanocomposites have more abundant deep traps (i.e., much higher trap density and energy level) compared to LDPE [31]. The introduction of ZnO nano-particles with low filler loading (<10 wt%) into LDPE results in a significant increase of deep trap density [26,32]. The trap distribution obtained from SPD results exhibit that the 5 wt% epoxy/TiO₂ nanocomposite presents the maximum value in trap energy level compared to pure epoxy and the 7 wt% epoxy/TiO₂ nanocomposite [33].

At higher filler loading, much shallower traps may be introduced in nanocomposites. From the trap distributions, the energy level of the 7 wt% epoxy/TiO₂ nanocomposite is a little lower than that of the 5 wt%, and this is considered to be caused by the overlap of interaction zones [33]. On the other hand, the dc resistivity is closely related to traps. The smaller dc resistivity may be attributed to the formation of shallower traps. The LDPE/ZnO or epoxy/ZnO nanocomposites with higher filler loading usually present a decrease in resistivity [34, 35]. According to the multi-core model which consists of a bonded layer, a bound layer and a loose layer, the formation of shallower traps may be explained from two ways [28]. In one way, a new trap distribution may be added to the original

traps. As the loose layer is low in density and high in free volume, defects and impurities are likely to exist in this region, and some of them will act as traps. The energy levels of these traps are shallower than those of traps originally existed in polymer matrix. In another way, the new trap distribution replaced the original traps. The interaction zone, especially the loose layers, can accommodate the original traps. In addition, they can make the original traps shift to shallower levels because of the far-distance effect. Thus, shallower traps are easily formed in the 20 wt% LZN, and are similar to those in the 20 wt% LZM. However, for the same filler loading, it is well known that the volume fraction of the interaction zone in the nanocomposites is comparatively larger than that in the microcomposites, so the 20 wt% LZN presents a bit higher trap density of shallow traps and lower trap density of deep traps as well as lower in energy level of shallow and deep traps than the 20 wt% LZM.

6 CONCLUSION

In the paper, combining shallow and deep traps with detrapping mechanism after removing the applied field, a model was proposed to obtain the trap energy distribution in materials using SPD measurement. According to this proposed model, the SPD curves of positively and negatively charged LDPE, LDPE/ZnO microcomposite and nanocomposite were analyzed. The trap energy distributions of hole-type and electron-type traps in these materials were obtained. The unit of trap density obtained by the proposed model was m^{-3} . The trap energy distribution in materials was effectively obtained based on any assumed space charge distribution (i.e., uniform, linear and exponent distribution). The space charge distributions in the corona charged materials only have a little influence on the trap density in materials with the thickness of about 200 μm . Results also show that the shallow or deep trap distributions of hole-type and electron-type traps in any used material are different from each other. In addition, the LDPE/ZnO nanocomposite presents lower in energy level and higher in trap density of shallow traps as well as lower in trap density of deep traps, compared to LDPE and LDPE/ZnO microcomposite. These results may be closely related to the distinct nature of hole-type and electron-type traps as well as the special morphology of polymer composites.

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